Matthew Duke

MPhys Physics and Astrophysics Graduate

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First-Class MPhys Physics and Astrophysics graduate with a strong foundation in Python, numerical analysis and data-driven scientific research. Experienced in developing computational workflows, analysing complex datasets and applying statistical and modelling techniques to solve technical problems. Seeking to apply these quantitative and programming skills in data, modelling, software or engineering-focused graduate roles.

EDUCATION

2022 - 2026 – University of Sheffield
- MPhys Physics and Astrophysics – First-Class Honours

2016 - 2022 – St Benedict's Catholic School
- A-Level: Mathematics A*, Physics A, Chemistry A, Further Mathematics A, EPQ B
- Twelve GCSE's grades 8-7 including Maths, English, Physics, Chemistry, and Biology

KEY SKILLS

Programming: Python (intermediate), SQL (beginner)
Libraries: NumPy, Pandas, Matplotlib, scikit-learn
Analysis & Modelling: Numerical analysis, Monte Carlo methods, uncertainty propagation, statistical analysis, machine learning, model evaluation
Data: Data cleaning, validation, integration, visualisation and exploratory analysis
Software: Excel

TECHNICAL PROJECTS

MPhys Research Project – Census of O-Type Stars in the Small Magellanic Cloud
- Developed a Python-based analysis pipeline to integrate, validate and process data for over 450 stars from multiple astronomical catalogues.
- Applied numerical and statistical methods, including uncertainty propagation and Monte Carlo techniques, to derive physical properties and compare observations against theoretical models.
- Investigated inconsistencies and anomalies across the dataset, producing clear visualisations and reproducible documentation to support robust interpretation of results.

Independent Projects (Further details available at matthewduke.co.uk.)
- Developed and optimised supervised machine learning models using real-world data, applying scikit-learn, statistical analysis and model evaluation techniques.
- Building a structured football match dataset for statistical and predictive analysis, developing skills in SQL, data pipelines and predictive modelling.

EMPLOYMENT HISTORY

National Trust – Food & Beverage Team Member
June 2024 – Present
- Trusted to open, operate and close multiple food outlets, recognised by team leaders for reliability and flexibility.
- Worked across departments to deliver efficient customer service while maintaining food-safety standards.

Just Recruitment Group – Agency Worker
June 2023 – September 2023
- Worked across warehousing, distribution and factory environments, adapting quickly to different teams, sites and responsibilities.
- Demonstrated reliability and flexibility when taking on varied assignments at short notice.

Tesco – Customer Assistant
September 2020 – April 2021
- Supported checkout operations, stock replenishment, customer service and store security in a busy supermarket during the pandemic.
- Balanced regular employment with A-level studies, developing strong time-management and communication skills.

LEADERSHIP, ACHIEVEMENTS & DEVELOPMENT

- Vice-President, University of Sheffield Darts Society 2025/26: helped organise major events, secure sponsorship and support society operations; represented Sheffield in league, Varsity and national competitions.
- Continuing to develop technical skills through independent Python projects and SQL study.
- Mentored by an experienced FX and interest-rates professional, gaining exposure to financial markets, risk and data-driven decision-making.
- University Darts 2025/26 UK National Champion